Charmed-baryon production in antiproton-proton collisions within an effective Lagrangian model

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OUTLINE
Charmed baryon production

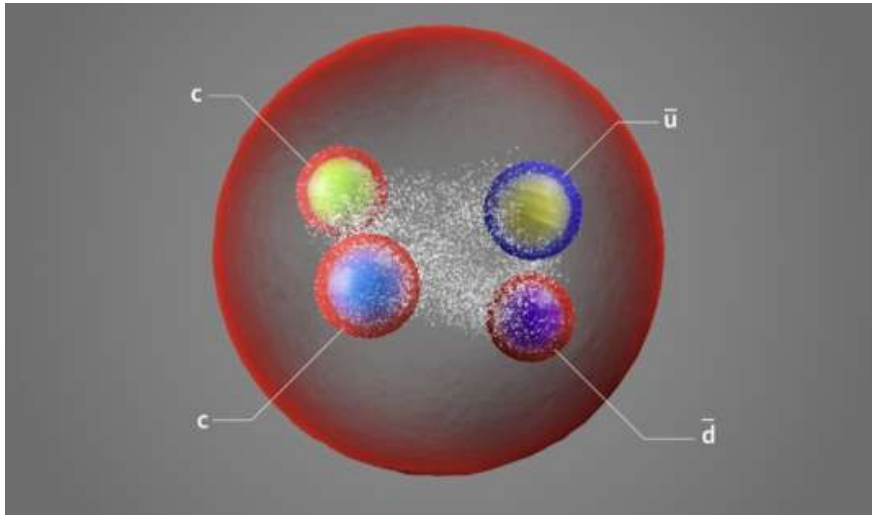


Fig. 1 An exotic tetraquark
[Ingrid Fadelli, Phys.org]

I. Introduction
- What is the Charmed baryon ?
- Why do we study charmed baryon production ?
- Current experiment

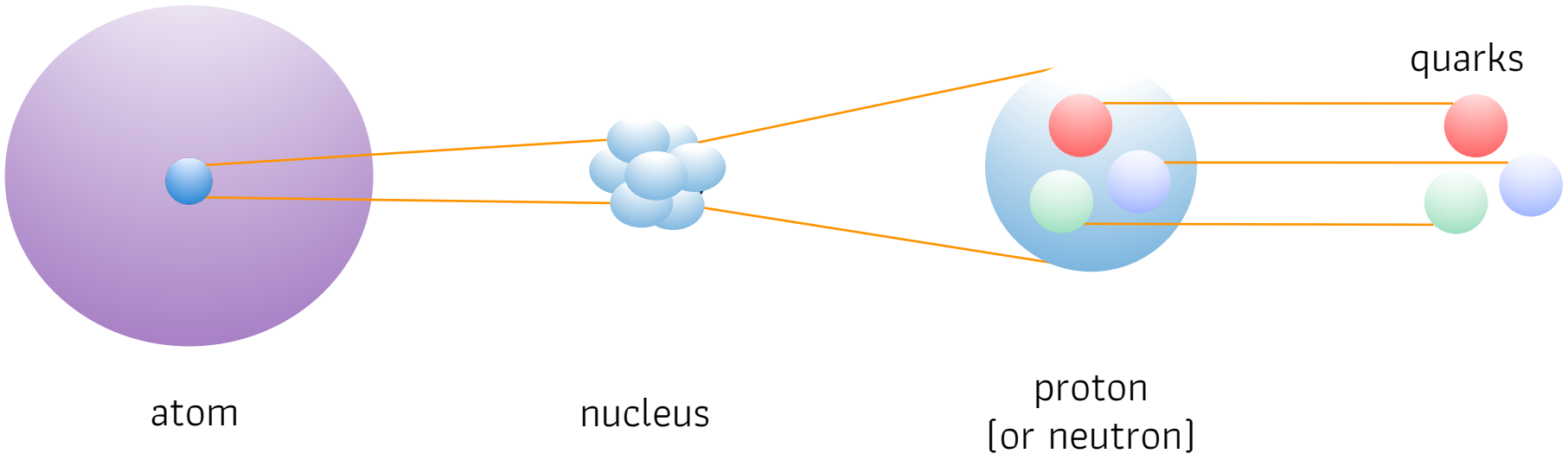
II. Model/Approach
- **Effective Lagrangain**

III. Results

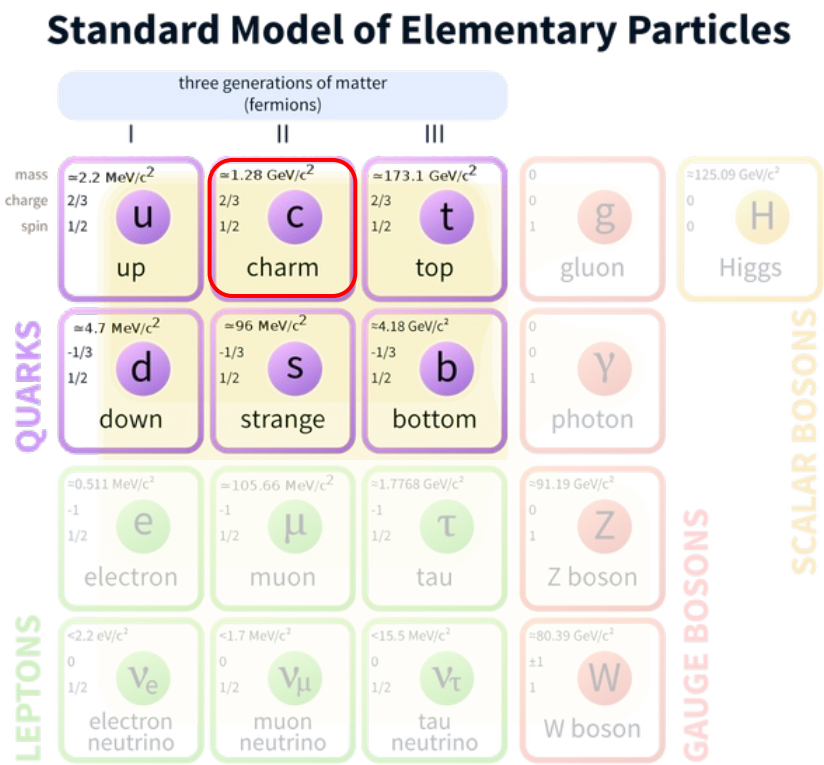
IV. Conclusion

INTRODUCTION

What is the “Charmed baryon”?



What is the “Charmed baryon”?



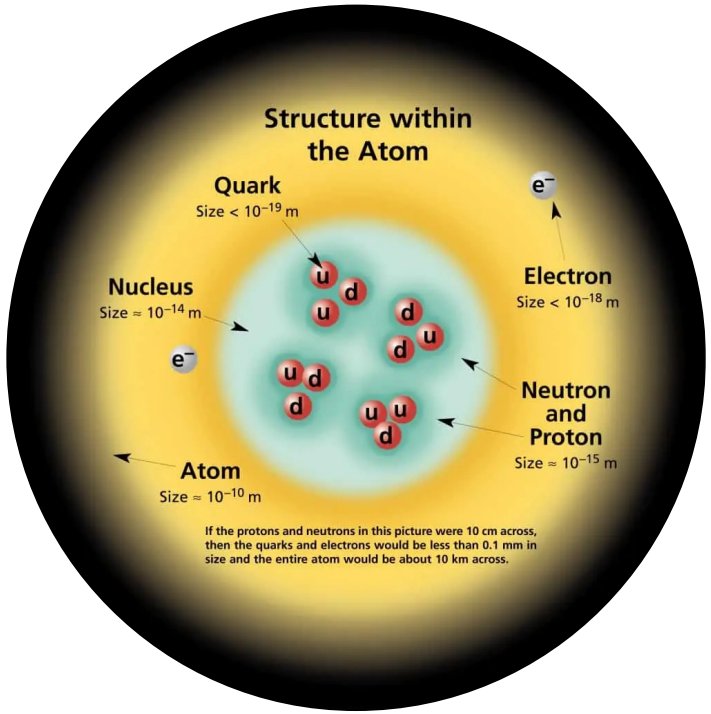


Fig.2

- Charm is one of the six 'flavors' of quarks, which are elementary particles.
- Baryons are heavy subatomic particles that are made up of three quarks.

Ref.[1] : https://medium.com/the-explanation/on-atoms-protons-neutrons-and-electrons-6347f40b1bdb

“Charmed baryon”

- **Composite particles** comprising all baryons made of at least one **charm quark.**
- The first charmed baryon states were detected in 1975 in neutrino interactions.

Notation	Content	J^P	SU(3)	I_3	S	C	Mass (GeV)
Λ_c^+	$c[ud]$	$1/2^+$	$\bar{3}$	0	0	1	2.286
Ξ_c^+	$c[su]$	$1/2^+$	$\bar{3}$	1/2	-1	1	2.466
Ξ_c^0	$c[sd]$	$1/2^+$	$\bar{3}$	-1/2	-1	1	2.472
Σ_c^{++}	cuu	$1/2^+$	6	1	0	1	2.453
Σ_c^+	$c\{ud\}$	$1/2^+$	6	0	0	1	2.451
Σ_c^0	cdd	$1/2^+$	6	-1	0	1	2.452

Λ_c^+ ← Charge
← Heavy quark (one charmed quark)

Ξ_c^+

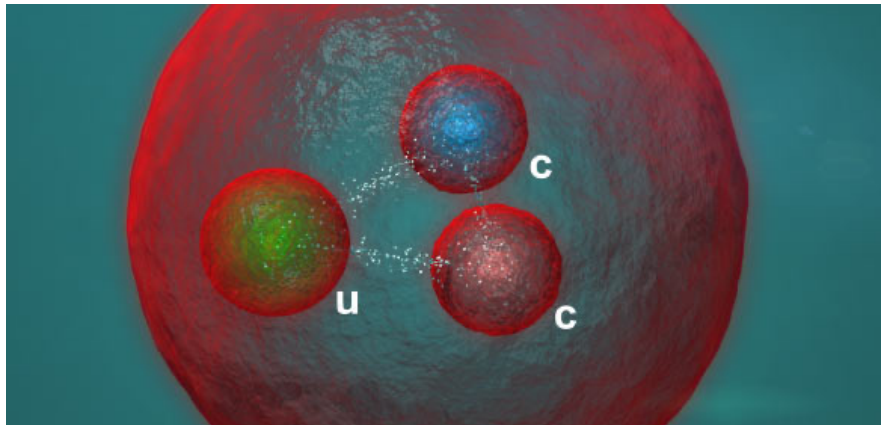


Fig. 3 The doubly charmed baryon Ξ_{cc}^{++} contains two charm quarks and one up quark. (Daniel Dominguez, CERN.)

➢ Important Properties

1. Mass
2. Lifetime
3. Intrinsic width
4. decay modes

INTRODUCTION

Why do we study charmed baryon?

➢ **Motivation**

- It provides better **understanding** of **Quantum Chromodynamics (QCD)** which is the fundamental theory of **the strong interaction.**
- Containing of a charm quark are quite distinct in their properties from the light flavored hadrons.

- However, the production and spectroscopy of the charmed baryons have not been explored in more detail.

The strong interaction.

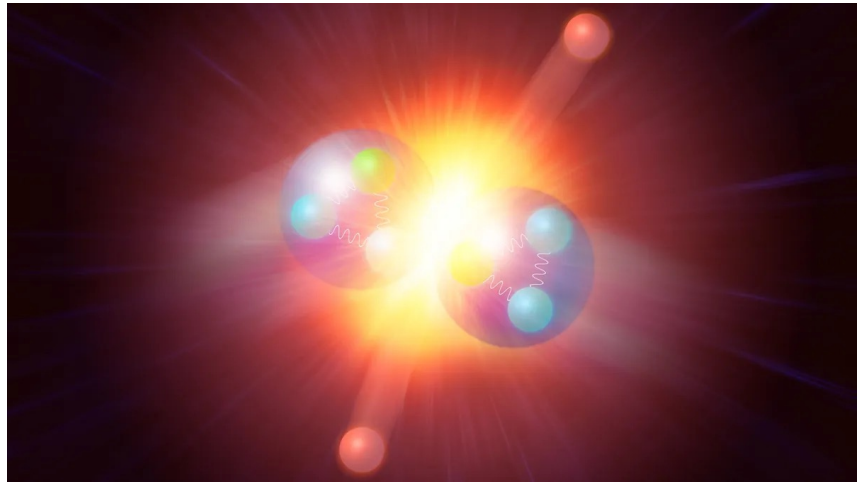


Fig. 4 The quarks are held together by the nuclear strong force carried by gluons.
[MARK GARLICK/SCIENCE PHOTO LIBRARY]

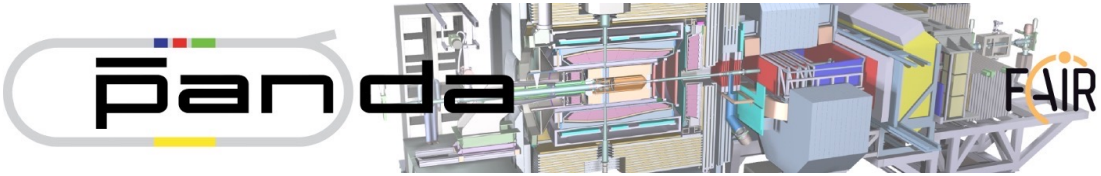
Current experiment

- **Most current experiments**

  Production of the ground state charmed baryon [Λc (2286)] by electron-positron annihilation experiments.

- **In the near future.**

  The studying in the proton-antiproton ($P\bar{P}$) annihilation at Darmstadt "($\bar{P}ANDA$)" experiment at the Facility for Antiproton and Ion Research (FAIR) in GSI, Darmstadt, Germany



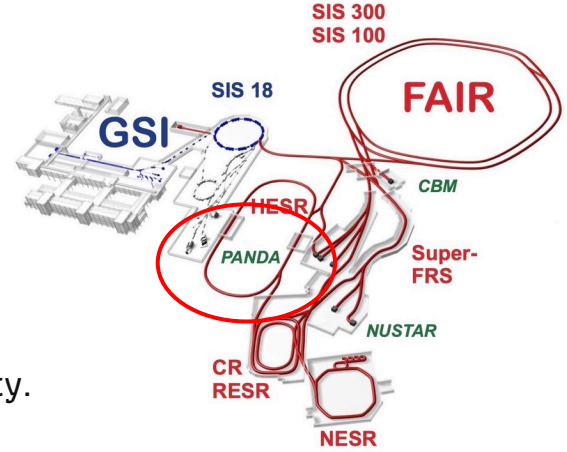


Fig 5. Overview of the future FAIR facility.

For the planning of this experiment, reliable theoretical estimates "the cross section" of this reaction would be of crucial importance.

INTRODUCTION

- **The cross section (σ) :** In particle physics, it means the probability that two particles will collide and react in a certain way.

- **Differential cross section** is defined to be :

$$\frac{d\sigma}{d\Omega}(\theta,\phi) = |f(\theta,\phi)|^2.$$

It is related to the scattering amplitude $|f(\theta,\phi)|^2$. That describes how the particle is scattered in various directions.

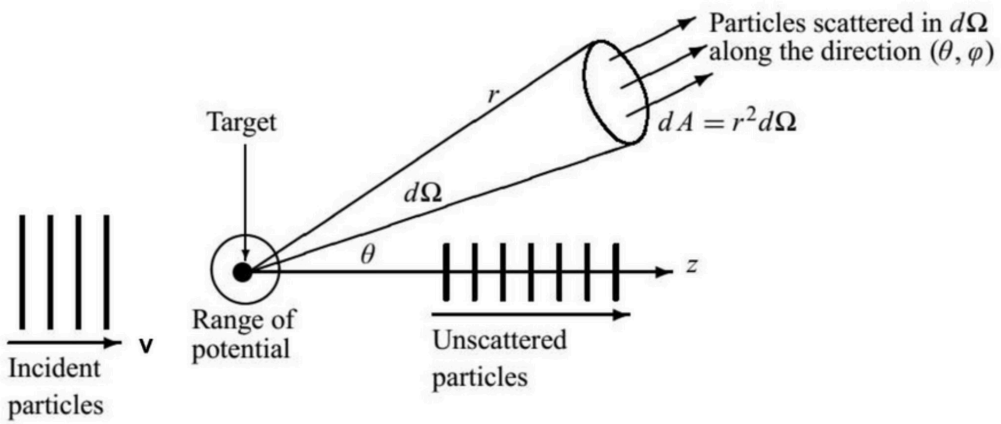


Fig. 6 The scattering between an incident beam of particles and a fixed target.
[John-Wiley, Chichester, 2001]

Charmed-baryon production in antiproton-proton collisions

$$P\overline{P} \rightarrow \overline{\Lambda}_c^- \Lambda_c^+$$

Model/Approach

➢ Effective Lagrangian

- It is a mathematical expression used in theoretical physics, particularly in quantum field theory.
- This approach serves as an approximation to describe the dynamics of a system or the behavior of fields and particles.
- Physicists use an effective Lagrangian that captures the essential features of the interactions at lower energies or over specific ranges.

Effective Lagrangian model

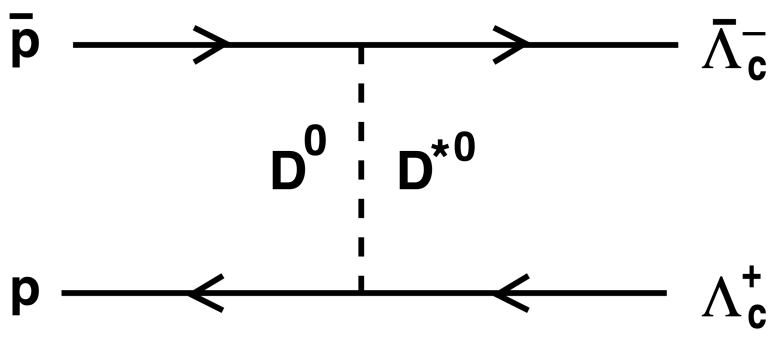


Fig. 7 model used to describe the $P\bar{P} \to \bar{\Lambda}_c^- \Lambda_c^+$ reaction.

01 FORMALISM

- Effective Lagrangians,
- The amplitudes
- The transition amplitude with distortion effects.

02 Results

R. Shyam[1] and H. Lenske[2]

arXiv:1406.7071v2 [hep-ph] 12 Jul 2014

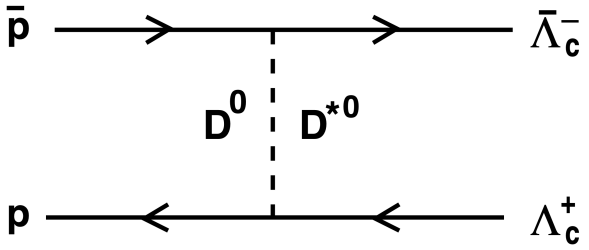
Effective Lagrangian model

01 FORMALISM

: To evaluate various amplitudes for the processes, Using the effective Lagrangians at the charm baryon-meson-nucleon vertices.

For the D^0 meson [mass [m_{D^0}] = 1.865 GeV] exchange vertices we have

$$\mathcal{L}_{D^0BN} = ig_{BD^0N}\bar{\psi}_B i\gamma_5\psi_N\phi_{D^0} + H.c.,$$



For the D^{*0} meson [mass [m_{D^0}] = 2.007 GeV] exchange vertices

$$\mathcal{L}_{D^{*0}BN} = g_{D^{*0}BN}\bar{\psi}_B\gamma_\mu\psi_N\theta^\mu_{D_{*0}} + \frac{f_{D^{*0}BN}}{4M}\bar{\psi}_B\sigma_{\mu\nu}\psi_N F^{\mu\nu}_{D^{*0}} + H.c.,$$

Effective Lagrangian model

01 FORMALISM

To evaluate various amplitudes for the processes.

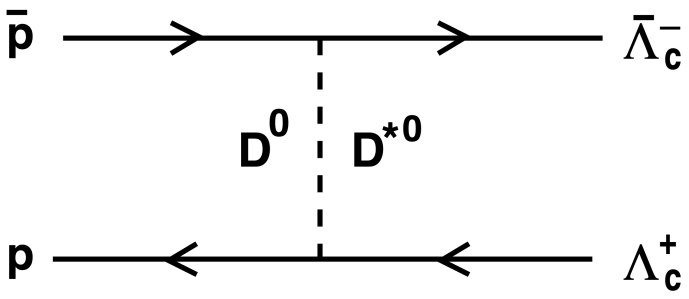


TABLE I. **Coupling constants** at the BD^0N and $BD^{*0}N$ vertices. These are taken from Ref. [2] where they are deduced from $DN$and $\bar{D}N$ scattering analysis.

Vertex	$g_{DBN}/\sqrt{4\pi}$	$f_{DBN}/\sqrt{4\pi}$
ND^0B	3.943	–
$ND^{*0}B$	1.590	5.183

Here B represents the charmed baryon.

[2] J. Haidenbauer, G. Krein, U.-G. Meissner, and L. Tolos, Eur. Phys. J. A **47**, 18 (2011).

Effective Lagrangian model

01 FORMALISM

To evaluate various amplitudes for the processes.

The **off-shell behavior** of the vertices is regulated by a monopole form factor:

$$F_i(q_i) = \frac{\lambda_i^2 - m_{D_i}^2}{\lambda_i^2 - q_{D_i}^2},$$

For calculating the amplitudes, propagators for the D^0 and D^{*0} mesons:

$$G_{D^0}(q) = \frac{i}{q^2 - m_{D^0}^2},$$

$$G_{D^{*0}}^{\mu\nu}(q) = -i\left(\frac{g^{\mu\nu} - q^\mu q^\nu/q^2}{q^2 - (m_{D^{*0}} - i\Gamma_{D^{*0}}/2)^2}\right).$$

$\Gamma_{D^{*0}}$ is the total width of the $D^{*0}[2007]$ meson ≈2.0 MeV.

- **Effective Lagrangians**,
  coupling constants, and forms of the propagators.
- **The amplitudes**
- **The transition amplitude** with distortion effects.

The strong sensitivity of the calculated cross sections to the distortion effects in the initial and final states.

$$T^{\bar{p}p\to\bar{\Lambda}_c^-\Lambda_c^+} = \sqrt{\Omega^{\bar{p}p}}\,T_{Born}^{\bar{p}p\to\bar{\Lambda}_c^-\Lambda_c^+}\,\sqrt{\Omega^{\bar{\Lambda}_c^-\Lambda_c^+}}$$

This work neglects the real part of the baryon-antibaryon interaction. Considering the $P\bar{P}$ initial state interaction, describing the strong absorption by an imaginary potential of Gaussian shape with range parameter μ and strength V_0. By using the **eikonal approximation**,

$$\Omega^{\bar{p}p} = exp[\frac{-\sqrt{\pi}EV_0}{\mu k}exp(-\mu^2 b^2)],$$

The parameters V_0 and μ may change with energy, Only three fixed parameters in this calculations.

02 Results and Discussions

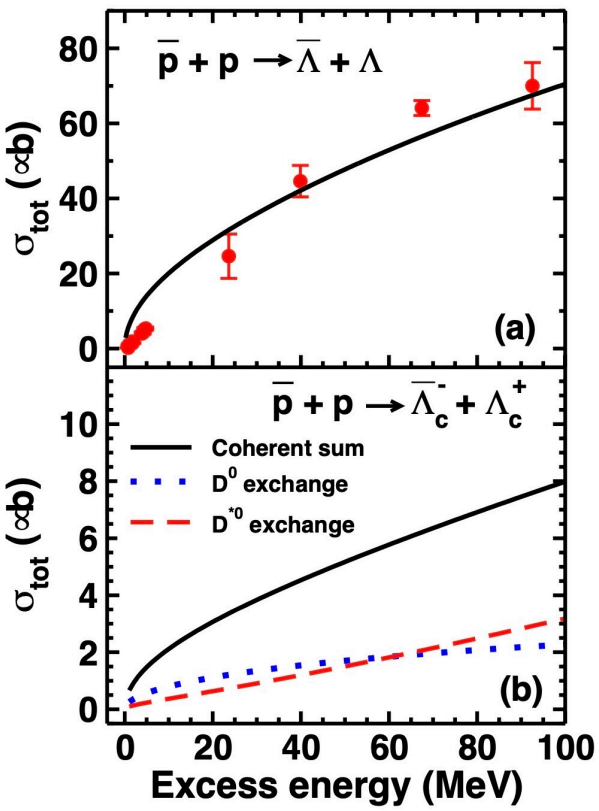


FIG. 1 Total cross section for the reactions as a function of the **excess energy**.

- In [a] the experimental data.
- In [b] the contributions of the D^0 and D^{*0} exchange processes.

02 Results and Discussions

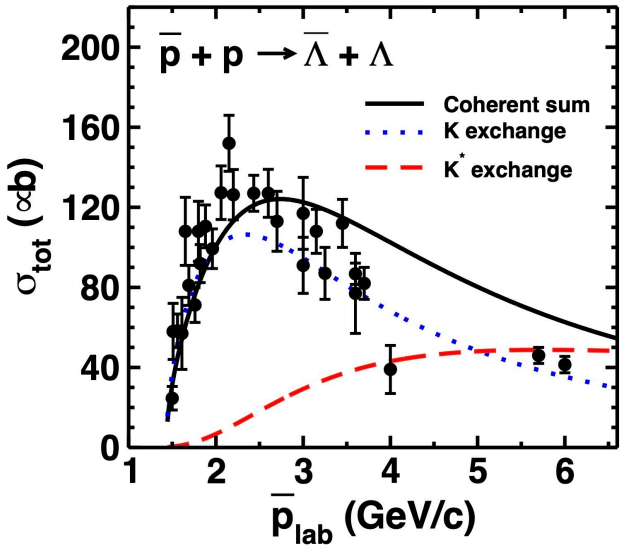


Fig. 2

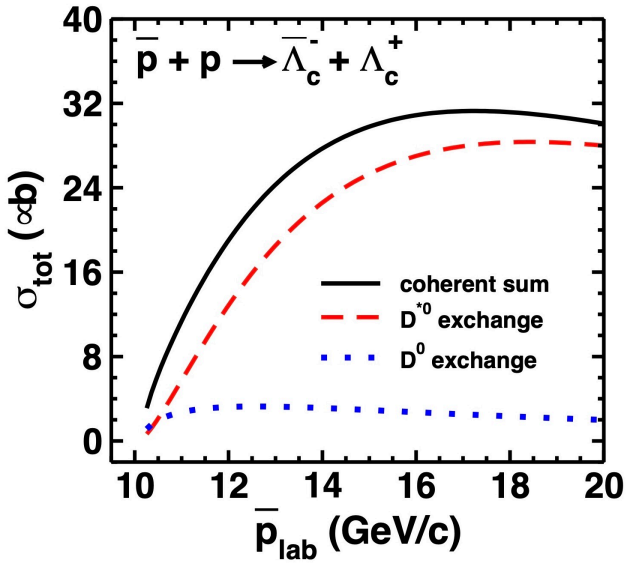


Fig. 3

Fig. 2 shows total cross section for the reaction as a function of the antiproton beam momentum. The individual contributions of $K[494]$ and $K^*[892]$ exchange processes. and Fig. 3 The contributions of the D^0 and D^{*0} exchange processes.

02 Results and Discussions

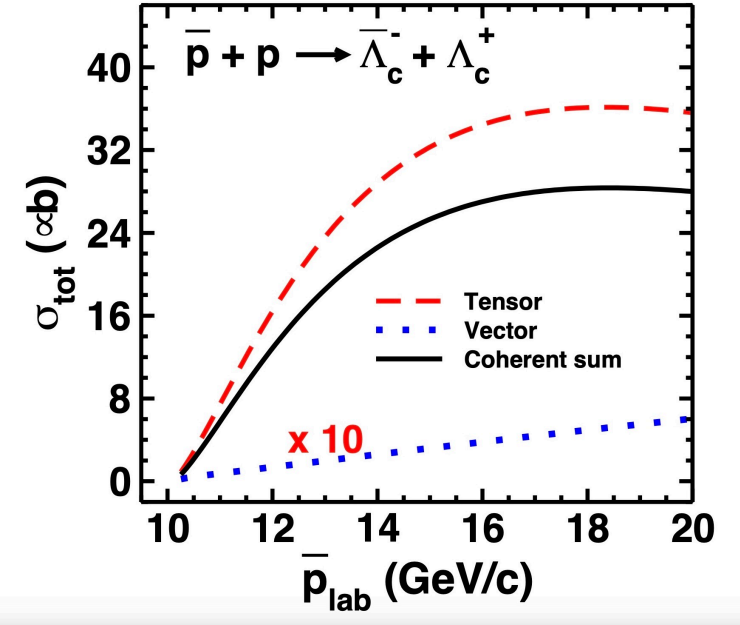


FIG. 5. Contributions of the vector and tensor coupling terms to the D^{*0} exchange as a function of the antiproton beam momentum.

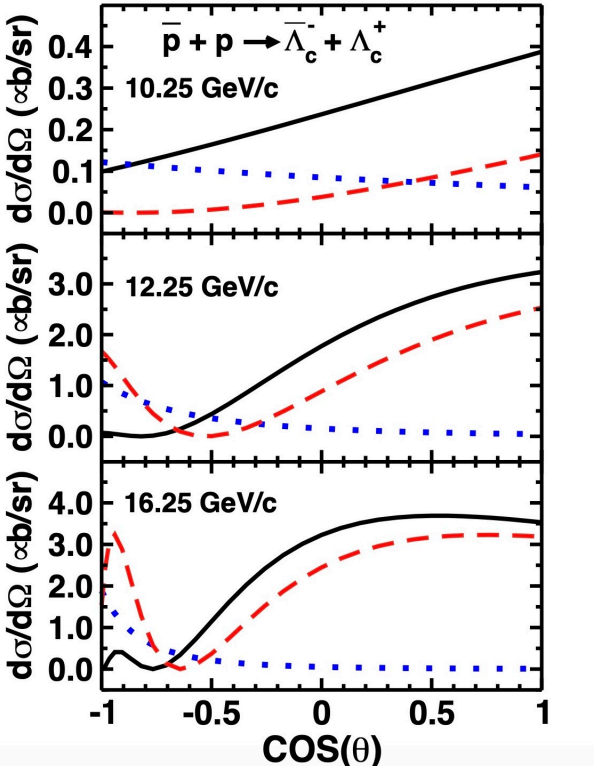


FIG. 6. Differential cross sections for the $P\bar{P} \rightarrow \bar{\Lambda}_c^- \Lambda_c^+$ reaction at the beam momenta of 10.25, 12.25, and 16.25 GeV/c. Contributions of the D^0 and D^{*0} exchange processes are shown by dotted and dashed lines, respectively,

Conclusion

- The total cross section for the $\boldsymbol{P\bar{P}} \rightarrow \bar{\boldsymbol{\Lambda}}_{\boldsymbol{c}}^{-}\boldsymbol{\Lambda}_{\boldsymbol{c}}^{+}$ vary between 1 and 8 μb at near-threshold beam momenta. This value agrees with that reported in other model calculations of Refs. [3,4]. For higher beam momenta, the cross sections are larger.

- The vector meson $[D^{*0}]$ exchange terms dominate the cross sections for all beam momenta except for those very close to the production threshold.

- The large cross sections obtained at higher beam momenta as compared to earlier studies are sturdy and can assist in strategizing experiments to observe this channel at the $\bar{P}ANDA$ facility.

[3] J. Haidenbauer, and G. Krein, Phys. Lett. B **678**, 314 (2010).
[4] J. Haidenbauer, and G. Krein, Few-Body Syst. **50**, 183 (2011).